Vector equations of lines and curves — Solutions

Warm-up

Question 1:
The vector equation of a line is given by $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, where $\mathbf{a}$ is a position vector of a point on the line and $\mathbf{d}$ is the direction vector.:$\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}\mathbf{d} = \mathbf{i} - \mathbf{j}$these into the general form:

$$\mathbf{r} = (2\mathbf{i} + 3\mathbf{j}) + \lambda(\mathbf{i} - \mathbf{j})$$

Question 2:
The vector equation of a line is $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$.the equation:$\mathbf{r} = \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$inspection, the position vector $\mathbf{a}$ is $\begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix}$, which corresponds to the point $(1, 5, -2)$.direction vector $\mathbf{d}$ is the vector multiplied by the parameter λ:

$$\mathbf{d} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$$

Question 3:
To find the vector equation of a line passing through two points $P(6, 7)$ and $Q(8, 9)$, we first find the direction vector $\mathbf{d} = \overrightarrow{PQ}$.

$$\begin{aligned} \mathbf{d} &= \begin{pmatrix} 8 \\ 9 \end{pmatrix} - \begin{pmatrix} 6 \\ 7 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 2 \end{pmatrix} \end{aligned}$$

the point $(6, 7)$ as the position vector $\mathbf{a} = \begin{pmatrix} 6 \\ 7 \end{pmatrix}$, the vector equation is:

$$\mathbf{r} = \begin{pmatrix} 6 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Question 4:
The vector equation of a circle with center $\mathbf{c}$ and radius R is given by $|\mathbf{r} - \mathbf{c}| = R$.the circle passes through the point $(1, 4)$, and the radius is 7, the center $\mathbf{c}$ must be at a distance of 7 from $(1, 4)$. However, the question asks for the equation of *a* circle. Assuming the point $(1, 4)$ is the center (which is a common interpretation if not specified otherwise, though the wording 'passing through' usually implies the point is on the circumference). If the point $(1, 4)$ is on the circumference, there are infinitely many such circles. Let's assume the center is at the origin $(0, 0)$ for simplicity, but that wouldn't fit the radius. Let's assume the center is $\mathbf{c} = (x_0, y_0)$ such that $|(1, 4) - (x_0, y_0)| = 7$.we assume the point $(1, 4)$ is the center, the equation is $|\mathbf{r} - (\mathbf{i} + 4\mathbf{j})| = 7$.

If the point is on the circumference, we can pick any center $\mathbf{c}$ such that the distance is 7. For example, let $\mathbf{c} = (8, 4)$.interpretation for this level: The point given is the center unless specified as 'passing through'. But the text says 'passing through'. Let's provide the general form where $\mathbf{c}$ is the center:

$$|\mathbf{r} - \mathbf{c}| = 7, \text{ where } |\mathbf{c} - (\mathbf{i} + 4\mathbf{j})| = 7$$

Question 5:
Similar to the circle, the vector equation of a sphere with center $\mathbf{c}$ and radius R is $|\mathbf{r}-\mathbf{c}| = R$.radius $R = 7$ and the sphere passes through $(1, 4, 5)$.the center be $\mathbf{c}$. The condition that the sphere passes through the point is:

$$\left|\begin{pmatrix}1\\4\\5\end{pmatrix} - \mathbf{c}\right| = 7$$

vector equation of the sphere is:

$$|\mathbf{r} - \mathbf{c}| = 7$$

Question 6:
Given the parametric equations:$\mathbf{r} = (4\cos(t)-5)\hat{\imath}+(4\sin(t)+6)\hat{\jmath}$gives the parametric components:

$$x = 4\cos(t) - 5$$
$$y = 4\sin(t) + 6$$

the trigonometric terms:

$$x + 5 = 4\cos(t)$$
$$y - 6 = 4\sin(t)$$

both equations and add them to use the identity $\cos^2(t) + \sin^2(t) = 1$:

$$(x + 5)^2 = 16\cos^2(t)$$
$$(y - 6)^2 = 16\sin^2(t)$$

$$(x + 5)^2 + (y - 6)^2 = 16(\cos^2(t) + \sin^2(t))$$
$$(x + 5)^2 + (y - 6)^2 = 16(1)$$
$$(x + 5)^2 + (y - 6)^2 = 16$$

Skill-building

Question 1:
To find the intersection point, we set the components of $\mathbf{r_1}$ and $\mathbf{r_2}$ equal to each other.

$$\begin{pmatrix} 1+2\lambda \\ 4-8\lambda \\ 9+12\lambda \end{pmatrix} = \begin{pmatrix} 6+8\mu \\ 2-6\mu \\ u+15\mu \end{pmatrix}$$

gives a system of three equations:

$$\begin{aligned} 1+2\lambda &= 6+8\mu \quad (1) \\ 4-8\lambda &= 2-6\mu \quad (2) \\ 9+12\lambda &= u+15\mu \quad (3) \end{aligned}$$

(1) and (2) for λ and μ:

$$\begin{aligned} \text{From (1): } & 2\lambda - 8\mu = 5 \\ \text{From (2): } & -8\lambda + 6\mu = -2 \end{aligned}$$

the first by 4:

$$\begin{aligned} 8\lambda - 32\mu &= 20 \\ -8\lambda + 6\mu &= -2 \end{aligned}$$

the equations:

$$\begin{aligned} -26\mu &= 18 \\ \mu &= -\frac{18}{26} = -\frac{9}{13} \end{aligned}$$

μ back into $2\lambda - 8\mu = 5$:

$$\begin{aligned} 2\lambda - 8(-\frac{9}{13}) &= 5 \\ 2\lambda + \frac{72}{13} &= 5 \\ 2\lambda &= \frac{65-72}{13} \\ 2\lambda &= -\frac{7}{13} \\ \lambda &= -\frac{7}{26} \end{aligned}$$

find the intersection point using $\mathbf{r_1}$:

$$\begin{aligned} x &= 1+2(-\frac{7}{26}) = 1-\frac{7}{13} = \frac{6}{13} \\ y &= 4-8(-\frac{7}{26}) = 4+\frac{28}{13} = \frac{52+28}{13} = \frac{80}{13} \\ z &= 9+12(-\frac{7}{26}) = 9-\frac{42}{13} = \frac{117-42}{13} = \frac{75}{13} \end{aligned}$$

, find u using equation (3):

$$\begin{aligned} 9+12(-\frac{7}{26}) &= u+15(-\frac{9}{13}) \\ \frac{75}{13} &= u-\frac{135}{13} \\ u &= \frac{75+135}{13} = \frac{210}{13} \end{aligned}$$

Question 2:
Two lines in 2D never intersect if they are parallel and not coincident.vector of $\mathbf{r_1}$ is $\mathbf{d_1} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$.vector of $\mathbf{r_2}$ is $\mathbf{d_2} = \begin{pmatrix} 1 \\ m \end{pmatrix}$.the lines to be parallel, $\mathbf{d_1}$ must be a scalar multiple of $\mathbf{d_2}$:

$$\begin{pmatrix} 2 \\ 8 \end{pmatrix} = k \begin{pmatrix} 1 \\ m \end{pmatrix}$$

components:

$$\begin{aligned} 2 = k(1) \implies k &= 2 \\ 8 = k(m) \implies 8 &= 2m \implies m = 4 \end{aligned}$$

if they are coincident: If $m = 4$, $\mathbf{r_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 8 \end{pmatrix}$ and $\mathbf{r_2} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 4 \end{pmatrix}$.$(2,1)$ from $\mathbf{r_2}$ is on $\mathbf{r_1}$ if:

$$\begin{aligned} 2 = 1 + 2\lambda \implies \lambda = 0.5 \\ 1 = 0 + 8\lambda \implies \lambda = 0.125 \end{aligned}$$

λ is inconsistent, the lines are parallel and not coincident. Thus, $m = 4$.

Question 3:
From the previous question, the lines are:$\mathbf{r_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 8 \end{pmatrix}$ and $\mathbf{r_2} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ (with $m = 4$).$\mathbf{r_1}$:

$$\begin{aligned} x = 1 + 2\lambda \implies \lambda = \frac{x-1}{2} \\ y = 8\lambda \implies \lambda = \frac{y}{8} \end{aligned}$$

λ:

$$\begin{aligned} \frac{x-1}{2} &= \frac{y}{8} \\ 4(x-1) &= y \\ y &= 4x - 4 \end{aligned}$$

$\mathbf{r_2}$:

$$\begin{aligned} x = 2 + \mu \implies \mu = x - 2 \\ y = 1 + 4\mu \implies \mu = \frac{y-1}{4} \end{aligned}$$

μ:

$$\begin{aligned} x - 2 &= \frac{y-1}{4} \\ 4x - 8 &= y - 1 \\ y &= 4x - 7 \end{aligned}$$

Question 4:
Given $\mathbf{r}(t) = (t^2 - 4)\mathbf{i} + (t^3 - t)\mathbf{j}$.with y-axis occurs when $x = 0$:

$$\begin{aligned} t^2 - 4 &= 0 \\ t^2 &= 4 \\ t &= \pm 2 \end{aligned}$$

y-coordinates for these t values:

$$\begin{aligned} \text{For } t = 2: &\quad y = 2^3 - 2 = 8 - 2 = 6 \\ \text{For } t = -2: &\quad y = (-2)^3 - (-2) = -8 + 2 = -6 \end{aligned}$$

of intersection are $(0, 6)$ and $(0, -6)$.find the Cartesian equation, eliminate t:

$$\begin{aligned} x &= t^2 - 4 \implies t^2 = x + 4 \\ y &= t(t^2 - 1) \end{aligned}$$

$t^2 = x + 4$ into the y equation:

$$\begin{aligned} y &= t(x + 4 - 1) \\ y &= t(x + 3) \\ y^2 &= t^2(x + 3)^2 \\ y^2 &= (x + 4)(x + 3)^2 \end{aligned}$$

Question 5:
a) Given $\mathbf{r}(t) = 3\cos(2t)\mathbf{i} + 3\sin(2t)\mathbf{j}$.equations are $x = 3\cos(2t)$ and $y = 3\sin(2t)$.

$$\begin{aligned} x^2 + y^2 &= (3\cos(2t))^2 + (3\sin(2t))^2 \\ x^2 + y^2 &= 9\cos^2(2t) + 9\sin^2(2t) \\ x^2 + y^2 &= 9(\cos^2(2t) + \sin^2(2t)) \\ x^2 + y^2 &= 9(1) \\ x^2 + y^2 &= 9 \end{aligned}$$

is the equation of a circle centered at the origin with radius $R = \sqrt{9} = 3$.b) Compare $\mathbf{r}(t) = 3\cos(2t)\mathbf{i} + 3\sin(2t)\mathbf{j}$ and $\mathbf{r}(u) = 3\cos(u)\mathbf{i} - 3\sin(u)\mathbf{j}$.$\mathbf{r}(t)$, the particle moves counter-clockwise around the circle. The angular velocity is 2 rad/s, meaning it completes a full revolution in $T = \frac{2\pi}{2} = \pi$ seconds.$\mathbf{r}(u)$, the particle moves clockwise (due to the $-\sin(u)$ term) with an angular velocity of 1 rad/s, completing a full revolution in $T = \frac{2\pi}{1} = 2\pi$ seconds.: $\mathbf{r}(t)$ moves twice as fast and in the opposite direction (counter-clockwise) compared to $\mathbf{r}(u)$.

Question 6:
a) Given $x^2 + y^2 + z^2 - 4x + 6y - 8z = 0$.the square for each variable:

$$\begin{aligned} (x^2 - 4x) + (y^2 + 6y) + (z^2 - 8z) &= 0 \\ (x - 2)^2 - 4 + (y + 3)^2 - 9 + (z - 4)^2 - 16 &= 0 \\ (x - 2)^2 + (y + 3)^2 + (z - 4)^2 &= 4 + 9 + 16 \\ (x - 2)^2 + (y + 3)^2 + (z - 4)^2 &= 29 \end{aligned}$$

vector form, this is $|\mathbf{r} - \mathbf{c}| = a$, where $\mathbf{c} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$ and $a = \sqrt{29}$.

$$\left| \mathbf{r} - \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \right| = \sqrt{29}$$

) The center is $(2, -3, 4)$ and the radius is $\sqrt{29}$.

Easier Exam Questions

Question 1:

The line is $\mathbf{r}_1(t) = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix} + t \begin{pmatrix} \lambda \\ 1 \\ 2 \end{pmatrix}$ and the sphere is $\left| \mathbf{r}_2 - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \right| = 3$.the line equation into the sphere equation:

$$\left| \begin{pmatrix} 4 + \lambda t \\ -2 + t \\ 2t \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \right| = 3$$
$$\left| \begin{pmatrix} 3 + \lambda t \\ t \\ 2t - 3 \end{pmatrix} \right| = 3$$

both sides:

$$(3 + \lambda t)^2 + t^2 + (2t - 3)^2 = 3^2$$
$$(9 + 6\lambda t + \lambda^2 t^2) + t^2 + (4t^2 - 12t + 9) = 9$$
$$\lambda^2 t^2 + 6\lambda t + 5t^2 - 12t + 9 = 0$$

as a quadratic in t:

$$(\lambda^2 + 5)t^2 + (6\lambda - 12)t + 9 = 0$$

the line to intersect the sphere at exactly one point, the discriminant must be zero ($\Delta = 0$):

$$\begin{aligned} \Delta &= (6\lambda - 12)^2 - 4(\lambda^2 + 5)(9) \\ 0 &= 36(\lambda - 2)^2 - 36(\lambda^2 + 5) \\ 0 &= (\lambda - 2)^2 - (\lambda^2 + 5) \\ 0 &= \lambda^2 - 4\lambda + 4 - \lambda^2 - 5 \\ 0 &= -4\lambda - 1 \\ 4\lambda &= -1 \\ \lambda &= -\frac{1}{4} \end{aligned}$$

option: (A).

Question 2:
Given $x = -|2\cos t|$ and $y = |2\sin t|$.that $x \leq 0$ and $y \geq 0$ for all t. This means the graph is restricted to the second quadrant.the relationship between x and y by squaring both:

$$x^2 = (-|2\cos t|)^2 = 4\cos^2 t$$
$$y^2 = (|2\sin t|)^2 = 4\sin^2 t$$

them together:

$$x^2 + y^2 = 4(\cos^2 t + \sin^2 t)$$
$$x^2 + y^2 = 4$$

is a circle of radius 2 centered at the origin. However, since $x \leq 0$ and $y \geq 0$, the graph is only the arc of the circle in the second quadrant., the correct graph is the one showing a quarter-circle in the second quadrant.

Question 3:

Points $A(1,0,2)$ and $B(3,9,6)$.vector $\mathbf{a} = \begin{pmatrix}1\\0\\2\end{pmatrix}$.vector $\mathbf{d} = \overrightarrow{AB} = \begin{pmatrix}3\\9\\6\end{pmatrix} - \begin{pmatrix}1\\0\\2\end{pmatrix} = \begin{pmatrix}2\\9\\4\end{pmatrix}$.equation of the line:

$$\mathbf{r} = \begin{pmatrix}1\\0\\2\end{pmatrix} + \lambda\begin{pmatrix}2\\9\\4\end{pmatrix}$$
$$\mathbf{r} = \mathbf{i} + 2\mathbf{k} + \lambda(2\mathbf{i} + 9\mathbf{j} + 4\mathbf{k})$$

option: (A).

Question 4:

Point $(-1, a)$ lies on the line $\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}2\\-2\end{pmatrix} + \mu\begin{pmatrix}-3\\4\end{pmatrix}$.$x = -1$ into the x-component equation:

$$-1 = 2 - 3\mu$$
$$-3 = -3\mu$$
$$\mu = 1$$

substitute $\mu = 1$ into the y-component equation to find a:

$$a = -2 + 4\mu$$
$$a = -2 + 4(1)$$
$$a = 2$$

option: (D).

Question 5:

Sphere 1: $|\mathbf{r}_1| = 6$ (Center $C_1(0,0,0)$, Radius $R_1 = 6$).2: $\left|\mathbf{r}_2 - \begin{pmatrix}4\\0\\0\end{pmatrix}\right| = 3$ (Center $C_2(4,0,0)$, Radius $R_2 = 3$).intersection of two spheres is a circle. The center of this circle lies on the line connecting the centers of the spheres (the x-axis in this case).the center of the intersection circle be $P(x,0,0)$. The distance from P to any point on the intersection circle is the radius r of the intersection circle.any point Q on the intersection circle, Q is on both spheres:

$$\begin{aligned} x_Q^2 + y_Q^2 + z_Q^2 &= 6^2 = 36 \\ (x_Q - 4)^2 + y_Q^2 + z_Q^2 &= 3^2 = 9 \end{aligned}$$

the second equation from the first:

$$\begin{aligned} x_Q^2 - (x_Q - 4)^2 &= 36 - 9 \\ x_Q^2 - (x_Q^2 - 8x_Q + 16) &= 27 \\ 8x_Q - 16 &= 27 \\ 8x_Q &= 43 \\ x_Q &= \frac{43}{8} \end{aligned}$$

the center of the intersection circle is the projection of Q onto the line connecting the sphere centers, the center is $(\frac{43}{8}, 0, 0)$.option: (D).

Question 6:
Point P is on the sphere with center $O(0,0,0)$. The radius squared is the distance OP^2:

$$\begin{aligned} &R^2 = |\overrightarrow{OP}|^2 = 4^2 + 5^2 + 6^2 \\ &R^2 = 16 + 25 + 36 \\ &R^2 = 77 \end{aligned}$$

which point has a distance squared from the origin equal to 77:

$$\begin{aligned} &\text{For A:} \quad 2^2 + 5^2 + 7^2 = 4 + 25 + 49 = 78 \\ &\text{For B:} \quad 5^2 + 5^2 + 5^2 = 25 + 25 + 25 = 75 \\ &\text{For C:} \quad 2^2 + 3^2 + 8^2 = 4 + 9 + 64 = 77 \\ &\text{For D:} \quad 2^2 + 6^2 + 6^2 = 4 + 36 + 36 = 76 \end{aligned}$$

C satisfies the condition. Correct option: (C).

Harder Exam Questions

Question 1:

The point $(2, y, z)$ lies on the line $\mathbf{r} = \begin{pmatrix} 6 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -1 \\ 5 \end{pmatrix}$.the x-component:

$$\begin{aligned} 2 &= 6 + 4\lambda \\ -4 &= 4\lambda \\ \lambda &= -1 \end{aligned}$$

$\lambda = -1$ into the y and z components:

$$\begin{aligned} &y = 2 + (-1)(-1) \\ &y = 2 + 1 \\ &y = 3 \end{aligned}$$

$$z = 3 + (-1)(5)$$
$$z = 3 - 5$$
$$z = -2$$

Question 2:

i) For $l_1 : x = y = z$, the direction vector is $\mathbf{d_1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and it passes through $(0, 0, 0)$.$l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, the direction vector is $\mathbf{d_2} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.for parallelism:

$$\mathbf{d_1} \neq k\mathbf{d_2} \implies \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \neq k \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

no such k exists, the lines are not parallel.for intersection:

$$x = \lambda$$
$$y = \lambda$$
$$z = \lambda$$

$$x = t$$
$$y = 3 - 2t$$
$$z = 2 + t$$

components:

$$\lambda = t$$
$$\lambda = 3 - 2t$$
$$\lambda = 2 + t$$

the first and third equations: $t = 2 + t \implies 0 = 2$, which is impossible., the lines are non-parallel and non-intersecting (skew lines).) The angle θ between the lines is the angle between their direction vectors $\mathbf{d_1}$ and $\mathbf{d_2}$:

$$\cos\theta = \frac{\mathbf{d_1} \cdot \mathbf{d_2}}{|\mathbf{d_1}||\mathbf{d_2}|}$$
$$\cos\theta = \frac{(1)(1) + (1)(-2) + (1)(1)}{\sqrt{1^2 + 1^2 + 1^2}\sqrt{1^2 + (-2)^2 + 1^2}}$$
$$\cos\theta = \frac{1 - 2 + 1}{\sqrt{3}\sqrt{6}}$$
$$\cos\theta = 0$$

$\cos\theta = 0$, the angle is $\theta = 90°$.

Question 3:
For lines L and M to be perpendicular, their direction vectors must have a dot product of zero.vector of L: $\mathbf{d_L} = \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix}$.vector of M: $\mathbf{d_M} = \begin{pmatrix} p \\ 2 \\ 2 \end{pmatrix}$.

$$\begin{aligned} \mathbf{d_L} \cdot \mathbf{d_M} &= 0 \\ (-2)(p) + (1)(2) + (-4)(2) &= 0 \\ -2p + 2 - 8 &= 0 \\ -2p - 6 &= 0 \\ -2p &= 6 \\ p &= -3 \end{aligned}$$

Question 4:
i) In a rectangular prism, adjacent edges are perpendicular. $\overrightarrow{OA}$ and $\overrightarrow{OC}$ are edges meeting at O.

$$\begin{aligned} \overrightarrow{OA} \cdot \overrightarrow{OC} &= 0 \\ (3\mathbf{i} - 12\mathbf{j} + 3\mathbf{k}) \cdot (2\mathbf{i} + \mathbf{j} + a\mathbf{k}) &= 0 \\ (3)(2) + (-12)(1) + (3)(a) &= 0 \\ 6 - 12 + 3a &= 0 \\ 3a - 6 = 0 \\ 3a &= 6 \\ a &= 2 \end{aligned}$$

) $\overrightarrow{OG}$ is the diagonal of the prism. In a rectangular prism, $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$ where D is the fourth vertex of the base. However, based on the standard labeling of a rectangular prism $OABC - DEFG$, $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OE}$ is not quite right. Let's use the property that $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$ where $\overrightarrow{OD}$ is the third edge from O.$\overrightarrow{OD} = x\mathbf{i} + y'\mathbf{j} + z'\mathbf{k}$. Since $\overrightarrow{OD}$ is perpendicular to $\overrightarrow{OA}$ and $\overrightarrow{OC}$:

$$\begin{aligned} \overrightarrow{OD} \cdot \overrightarrow{OA} = 0 \\ \overrightarrow{OD} \cdot \overrightarrow{OC} = 0 \end{aligned}$$

, $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$.

$$\begin{aligned} -2\mathbf{i} + y\mathbf{j} + z\mathbf{k} &= (3\mathbf{i} - 12\mathbf{j} + 3\mathbf{k}) + (2\mathbf{i} + \mathbf{j} + 2\mathbf{k}) + \overrightarrow{OD} \\ -2\mathbf{i} + y\mathbf{j} + z\mathbf{k} &= (5\mathbf{i} - 11\mathbf{j} + 5\mathbf{k}) + \overrightarrow{OD} \end{aligned}$$

, $\overrightarrow{OD} = (-2 - 5)\mathbf{i} + (y + 11)\mathbf{j} + (z - 5)\mathbf{k} = -7\mathbf{i} + (y + 11)\mathbf{j} + (z - 5)\mathbf{k}$.use $\overrightarrow{OD} \cdot \overrightarrow{OA} = 0$:

$$\begin{aligned} (-7)(3) + (y + 11)(-12) + (z - 5)(3) &= 0 \\ -21 - 12y - 132 + 3z - 15 &= 0 \\ -12y + 3z - 168 &= 0 \\ -4y + z &= 56(1) \end{aligned}$$

use $\overrightarrow{OD} \cdot \overrightarrow{OC} = 0$:

$$\begin{aligned}(-7)(2) + (y + 11)(1) + (z - 5)(2) &= 0\\ -14 + y + 11 + 2z - 10 &= 0\\ y + 2z - 13 &= 0\\ y + 2z &= 13 (2)\end{aligned}$$

the system (1) and (2):

$$\begin{aligned}z &= 4y + 56\\ y + 2(4y + 56) &= 13\\ y + 8y + 112 &= 13\\ 9y &= -99\\ y &= -11\end{aligned}$$

, the question asks to show $y = 0$ and $z = 2$. Let's re-read the prism geometry. If O is a vertex and A, C, D are adjacent vertices, then $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$. If the prism is $OABC - DEFG$, then G is the opposite vertex to O. $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$. Let's check the provided values $y = 0, z = 2$ in $\overrightarrow{OD} = \overrightarrow{OG} - \overrightarrow{OA} - \overrightarrow{OC}$.

$$\overrightarrow{OD} = \begin{pmatrix}-2\\0\\2\end{pmatrix} - \begin{pmatrix}3\\-12\\3\end{pmatrix} - \begin{pmatrix}2\\1\\2\end{pmatrix}$$
$$\overrightarrow{OD} = \begin{pmatrix}-7\\11\\-3\end{pmatrix}$$

if $\overrightarrow{OD} \perp \overrightarrow{OA}$:

$$\begin{aligned}\overrightarrow{OD} \cdot \overrightarrow{OA} &= (-7)(3) + (11)(-12) + (-3)(3)\\ &= -21 - 132 - 9 = -162 \neq 0\end{aligned}$$

is a contradiction in the problem statement's given values for y and z or the vector $\overrightarrow{OA}$. Let's assume the intended geometry was $\overrightarrow{OG} = \overrightarrow{OA} + \overrightarrow{OC} + \overrightarrow{OD}$ and solve for y, z based on the provided $\overrightarrow{OA}$ and $\overrightarrow{OC}$. However, since the prompt asks to 'show that', I will provide the logic for the intended result if the vectors were consistent.) Point E is the vertex above O. $\overrightarrow{OE} = \overrightarrow{OD}$. The height above the x-y plane is the z-component of $\overrightarrow{OE}$.

$$\begin{aligned}\text{Height} &= z_{OD}\\ &= z_G - z_A - z_C\\ &= 2 - 3 - 2 = -3\end{aligned}$$

height is 3 units.